QUANTUM COMPUTING AND MODERN PHYSICS
(For all IT Branches and RAI)

Course Code:	**PHY102**	**Course Type:**	**BSC**
Teaching Hours/Week (L:T:P: S):	**3:0:2:0**	**Credits:**	**04**
Total Teaching Hours:	**45+0+30**	**CIE + SEE Marks:**	**50+50**

Teaching Department: Physics

Course Objectives:

1.	To understand the concepts of quantum mechanics.
2.	To study the principles of quantum computing and its applications.
3.	To study the properties of conductors and superconductors and their applications.
4.	To identify the fundamental principles of semiconductors to apply in electronic devices.
5.	To understand the principles, construction, and applications of LASERs and Optical fibres.

UNIT-I

Quantum Mechanics — **09 Hours**

Introduction to quantum mechanics, de Broglie Hypothesis and Matter Waves, de Broglie wavelength, Phase Velocity and Group Velocity, Relation between Phase Velocity and Group Velocity (Derivation), Relation between Particle Velocity and Group Velocity, Heisenberg's Uncertainty Principle - Application of Uncertainty Principle (nonexistence of electron inside the nucleus),

Wave Function, Physical Significance of a wave function, Time independent Schrödinger wave equation (Derivation), Eigen functions and Eigen Values, Particle inside one dimensional infinite potential well, Numerical Problems.

UNIT-II

Quantum Computing — **09 Hours**

Introduction to Quantum Computing, Moore's law and and its end, Differences between Classical & and Quantum computing. Dirac representation and matrix operations: Matrix representation of 0 and 1 States, Identity Operator I, Applying I to $|0\rangle$ and $|1\rangle$ states, Row and Column Matrices and their multiplication (Inner Product), Orthogonality, Orthonormality. Concept of qubit and its properties. Representation of qubit by Bloch sphere. Single and Two qubits. Extension to N qubits. Numerical Problems.

Quantum Gates: Single Qubit Gates: Quantum Not Gate, Pauli – X, Y and Z Gates, Hadamard Gate, Phase Gate (or S Gate).

UNIT-III

Conductors and Superconductors	09 Hours
Free-electron concept. Classical free-electron theory – Assumptions, Drift velocity, mean collision time, mean free path and relaxation time, Expression for electrical conductivity in metals (Drude-Lorentz theory), Failures of classical free-electron theory. Quantum free -electron theory – Assumptions, Fermi energy and Fermi factor for metals and its variation with temperature, Numerical problems. Introduction to superconductors, characteristic properties: Meissner effect, Critical field, Critical Current. Type-I and Type-II superconductors. BCS theory (qualitative). High temperature superconductors, Numerical problems.	
UNIT-IV	
Semiconductors	**09 Hours**
Introduction to semiconductors, intrinsic and extrinsic semiconductors - carrier generation, Direct and indirect band gap semiconductors. Expression for concentration of electrons in conduction band and and holes concentration in valance band (only mention the expression), Fermi level in Intrinsic and Extrinsic Semiconductors and its behavior with temperature, Electrical conductivity of an intrinsic semiconductor (derivation) and effect of temperature on intrinsic conductivity, Hall effect, Expression for Hall coefficient (derivation) and its application. p-n junction: Junction formation, Unbiased and biased p-n junction, Devices: LED, Photodiode and solar cell. Numerical problems.	
UNIT-V	
LASER and Optical Fibers	**09 Hours**
Introduction to lasers, Characteristics of LASER, Interaction of radiation with matter, Einstein's coefficients (Qualitative), Requisites of a Laser System. Conditions for Laser action. Principle, Construction and Working of Nd:YAG LASER and semiconductor LASER, Application of Lasers in Bar code scanner and Laser Printer. Numerical Problems. Introduction and Principle of Optical Fibers (TIR), Propagation mechanism in optical fibers - Angle of Acceptance and Numerical Aperture N.A.), Expression for NA, Fractional Index Change, Modes of Propagation, Number of Modes and V Number, Types of Optical Fibers, Attenuation, Factors contributing and Mention of Expression for Attenuation coefficient, Numerical problems.	

List of Experiments	
1.	Energy gap of a semiconductor (any method).
2.	Hall effect
3.	I-V characteristics of Zener diode
4.	Solar cell characteristics.
5.	Determination of wavelength by Laser diffraction.

6.	Diffraction of White light by optical grating
7.	Determination of acceptance angle and numerical aperture of the given Optical Fiber.
8.	Photo electric effect – Determination of the work function of the material of the emitter of a photocell.
9.	Photo-Diode characteristics
10.	LED characteristics and determination of Planck's Constant using LEDs.
11.	Transistor Characteristics.
12.	Determination of Fermi Energy of Copper.
13.	Resonance in LCR circuits.
14.	Simulation experiment.

Course Outcomes: At the end of the course students will be able to

1.	Apply the knowledge of quantum mechanics to analyse the physical properties exhibited by particles at sub-atomic level.
2.	Apply the basic principles of Quantum Mechanics in Quantum Computing.
3.	Apply knowledge of conductor and superconductor properties to engineering applications.
4.	Analyze the role of semiconductors in electronic devices, exploring the properties of semiconductors.
5.	Analyze the properties of optical waves in the phenomena of lasing action and signal propagation.

Course Outcomes Mapping with Program Outcomes & PSO

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	PSO↓		
↓ Course Outcomes														
PHY102-1.1	3	2						2			1			
PHY102-1.2	3	2						2			1			
PHY102-1.3	3	2						2			1			
PHY102-1.4	3	2						2			1			
PHY102-1.5	3	2						2			1			

1: Low 2: Medium 3: High

TEXTBOOKS:

1.	Parag K Lala, "Quantum Computing – A Beginner's Introduction", Indian Edition, McGraw Hill, Reprint 2020.
2.	B. G. Streetmann, "Solid State Electronic devices", 6th edition, Prentice Hall India Learning Private Limited.
3.	A. Ghatak, "Optics", Tata McGraw Hill Pub., 5th Edition, 2012.

REFERENCE BOOKS:

1.	Michael A. Nielsen & Isaac L. Chuang, "Quantum Computation and Quantum Information", Cambridge Universities Press, 2010 Edition.
2.	Vishal Sahani, "Quantum Computing", McGraw Hill Education, 2007 Edition.
3.	Maria Luisa Dalla Chiara, Roberto Giuntini, Roberto Leporini, Giuseppe Sergioli, "Quantum Computation and Logic: How Quantum Computers Have Inspired Logical Investigations", Trendsin Logic, Volume 48, Springer.
4.	Gupta and Kumar, "Solid State Physics", K. Nath & Co., Meerut.
5.	A. J. Dekker, "Electrical Engineering Materials", Prentice Hall India Pub., New Delhi, Reprint 2011.
6.	S. O. Pillai, "Solid State Physics", New Age International Private Limited, 8th Edition, 2018.
7.	M. Ali. Omar, "Elements of Solid State Physics: Principles and Applications", Pearson Publishers.
8.	Arthur Beiser, "Concepts of Modern Physics", Tata McGraw Hill Education Private Limited, Special Indian Edition, 2009.
9.	Kenneth Krane, "Modern Physics", Wiley International, 3rd Edition, 2012.
10.	Michael Tinkham, "Introduction to Superconductivity", II Edition, McGraw Hill, INC
E Books / MOOCs/ NPTEL/ Web links	
1.	LASER: https://www.youtube.com/watch?v=WgzynezPiyc
2.	Superconductivity: https://www.youtube.com/watch?v=MT5Xl5ppn48
3.	Optical Fiber: https://www.youtube.com/watch?v=N_kA8EpCUQo
4.	Quantum Mechanics: https://www.youtube.com/watch?v=p7bzE1E5PMY&t=136s
5.	Quantum Computing: https://www.youtube.com/watch?v=jHoEjvuPoB8
6.	Quantum Computing: https://www.youtube.com/watch?v=ZuvCUU2jD30
7.	Physics of Animation: https://www.youtube.com/watch?v=kj1kaA_8Fu4
8.	Statistical Physics Simulation: https://phet.colorado.edu/sims/html/plinko-probability/latest/plinkoprobability_en.html
9.	NPTEL Superconductivity: https://archive.nptel.ac.in/courses/115/103/115103108/
10.	NPTEL Quantum Computing: https://archive.nptel.ac.in/courses/115/101/115101092
11.	Virtual LAB: https://www.vlab.co.in/participating-institute-amrita-vishwa-vidyapeetham
12.	Virtual LAB: https://vlab.amrita.edu/index.php?sub=1&brch=189&sim=343&cnt=1
Activity Based Learning (Suggested Activities in Class)/ Practical Based learning	
1.	http://nptel.ac.in
2.	https://swayam.gov.in
3.	https://virtuallabs.merlot.org/vl_physics.html
4.	https://phet.colorado.edu
5.	https://www.myphysicslab.com